

Project Report on Symbolic Regression

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1 Introduction

- What is symbolic regression and why is it relevant?
- Why do standard ML methods struggle with extrapolation?
- What is happening in this paper (short overview).

2 Approach

- Describe the algorithm you built to discover symbolic formulas from data.
- Use mathematical notation! How do you represent expressions (e.g., expression trees, genetic programming, neural-guided search)?
- Cite properly! Related work includes PySR [1] or neural approaches to symbolic regression [2].
- Use pseudocode or figures to make difficult parts of your description understandable.

3 Experiments

Here you have freedom to find something interesting. Some possibilities would be:

- Compare different search strategies on the same benchmark functions.
- Ablation study: how does performance change with different operator sets or complexity penalties?
- Inspect your algorithm's extrapolation behaviour on functions of varying complexity.
- Test on real-world data where you expect an underlying symbolic relationship.

Get creative! Find appropriate representations that highlight your results (Tables, Plots).

4 Discussion

- Recap: What did you do?
- What are shortcomings/strengths of your approach?
- When does symbolic regression work well, and when does it fail?
- Discuss results and future directions.

References

- [1] Miles Cranmer. Interpretable machine learning for science with pysr and symbolicregression.jl. *CoRR*, abs/2305.01582, 2023.
- [2] Samuel Kim, Peter Y. Lu, Srijon Mukherjee, Michael Gilbert, Li Jing, Vladimir Ceperic, and Marin Soljacic. Integration of neural network-based symbolic regression in deep learning for scientific discovery. 2020.